

ERF Study (Task 5)

The Smooth (**erf**) Angular-Cost Hamiltonian for the TrackHHL Segment-Level Algorithm
Toy LHCb VELO Characterisation Study

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Abstract

This report characterises a *smooth* angular-compatibility cost for the TrackHHL 1-Bit Quantum Filter (1BQF) [1] segment-level Hamiltonian, replacing the hard step-function acceptance with the error-function (**erf**) kernel $C(\theta) = 1 + \text{erf}((\varepsilon - \theta)/(\theta_d\sqrt{2}))$, whose width θ_d is a tunable smoothing scale. On a fixed reference event ($T = 50$ tracks, five planes) we compare the step and **erf** Hamiltonians side by side: their matrix structure and eigenspectra, their classical and quantum solution vectors, and their segment-level efficiency / purity. Four findings are reported.

1. At the validated absolute operating threshold ($x > 0.35$), the step function is *already perfect* in low noise (efficiency = 1.00, false rate = 0.00); the **erf** kernel matches it but adds a small false-rate penalty ($\leq 2\%$) by giving partial weight to borderline-false pairs.
2. The **erf** kernel's advantage is *robustness to measurement resolution*: at $\sigma_{\text{res}} = 0.02$ mm the step efficiency collapses to 0.46–0.53 while **erf** ($\theta_d = 10^{-3}$) recovers it to 0.76–0.80, a +0.27 to +0.34 absolute gain.
3. The **erf** kernel roughly *doubles the quantum fidelity*: at $T = 10$ the cosine between the 1BQF and classical solutions rises from $\cos \theta_{QC} = 0.47$ (step) to 0.84 (**erf**, $\theta_d = 10^{-4}$), despite a larger matrix condition number.
4. The smoothing is noise-invariant in spectrum: the condition number κ depends only on θ_d ($\kappa_{\text{step}} = 2.4$, $\kappa_{\text{erf}} = 8.6$ –9.5) and is unchanged by hit drop.

This is the local, single-event portion of the study; the full HTCondor sweep over $\theta_d \times (\sigma_{\text{scatt}}, \sigma_{\text{res}}) \times T$ (882 jobs) is in progress and will extend the quantitative landscape.

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1 Introduction

The TrackHHL segment-level algorithm [1] reconstructs trajectories by solving the linear system $A\mathbf{x} = \mathbf{b}$, where each x_i is the activation of a hit-to-hit *segment* and the compatibility matrix A couples two segments sharing a middle hit when their kink angle θ is small. All previous tasks in this series use a *hard* acceptance: a triplet contributes a coupling of 1 if $\theta < \varepsilon$ and 0 otherwise. This report (Task 5) studies the only variant with a *soft* acceptance, in which the binary cut is replaced by a smooth **erf** kernel of tunable width θ_d . The step function is recovered as $\theta_d \rightarrow 0$, which doubles as a regression check.

The motivation is twofold. Physically, a soft kernel weights triplets by how compatible they are rather than accepting or rejecting them outright, which could improve tolerance to measurement smearing near the acceptance boundary. Algorithmically, the smoothing reshapes the spectrum of A , which directly affects the quantum (HHL) solver whose cost and fidelity depend on the eigenvalue distribution.

Section 2 defines both Hamiltonians and the scoring convention. Section 3 describes the fixed reference event and the parameter grid. Section 4 presents the structural, solution-vector, segment-metric, and quantum comparisons. Sections 5–6 interpret the results and state conclusions.

2 Theoretical Framework

2.1 The step and erf Hamiltonians

A triplet is two segments sharing a middle hit on consecutive planes, with kink angle $\theta = \arccos(\hat{v}_{\text{in}} \cdot \hat{v}_{\text{out}})$. The two compatibility kernels are

$$C_{\text{step}}(\theta) = \begin{cases} 1 & \theta < \varepsilon \\ 0 & \text{otherwise} \end{cases}, \quad C_{\text{erf}}(\theta) = 1 + \text{erf}\left(\frac{\varepsilon - \theta}{\theta_d \sqrt{2}}\right). \quad (1)$$

C_{erf} is a smooth sigmoid in $[0, 2]$ centred on $\theta = \varepsilon$ with transition width set by θ_d . As $\theta_d \rightarrow 0$ it approaches $2C_{\text{step}}$; for finite θ_d it assigns partial weight to triplets near the boundary. The off-diagonal entries of A are $-C(\theta)$ (attractive coupling), and the diagonal is the constant $\gamma + \delta = 4$ (with $\gamma = 3, \delta = 1$), independent of the kernel. The right-hand side is $\mathbf{b} = \delta \mathbf{1}$.

2.2 Matrix density

For the clean reference event ($T = 50$, $N_{\text{seg}} = 10\,000$) the step matrix has $\text{nnz} = 10\,300$ (essentially diagonal plus the ~ 150 true triplet couplings), whereas every **erf** matrix has $\text{nnz} = 760\,000$ — about $74\times$ denser — because all mid-hit pairs receive a (possibly tiny) non-zero weight. The **erf** kernel therefore trades sparsity for smoothness (Fig. 1).

2.3 The scoring convention

Segment metrics are evaluated by thresholding the solution vector at the *absolute* value $x > 0.35$, the validated convention from the segment-level baseline (`Verify_new_results/segment_level_analysis.ipynb`). Its justification is the Hopfield fixed-point structure of A : an isolated/false segment relaxes to $\delta/(\delta + \gamma) = 0.25$, while a true segment in a compatible chain sits on a ≈ 0.375 plateau, so 0.35 separates the two populations. A *relative* threshold $\tau \cdot \max(x)$ must *not* be used: it collapses below both levels and produces a spurious false rate near unity.

3 Study Design

The reference event is $T = 50$ pion tracks on five planes ($\Delta z = 33$ mm, $z_0 = 33$ mm, ± 40 mm active area, cone $|\phi|, |\theta| \leq 0.2$), generated with a fixed seed so that the step and **erf** Hamiltonians see identical hits. The smoothing width is swept over $\theta_d \in \{10^{-6}, 10^{-5}, 5 \times 10^{-5}, 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$, with $\theta_d = 10^{-6}$ serving as the step-function regression point. Three experimental sections are reported: (i) a perfectly clean event ($\sigma_{\text{scatt}} = \sigma_{\text{res}} = 0$, no hit drop); (ii) the same event with a 1% hit-drop noise; and (iii) a scan over $\sigma_{\text{scatt}} \in \{0, 10^{-4}, 5 \times 10^{-4}\}$ and $\sigma_{\text{res}} \in \{0, 5 \times 10^{-6}, 0.02\}$ mm. Eigenspectra are computed on a smaller $T = 20$ event (dense **eigvalsh**); the quantum comparison uses $T = 10$ ($N_{\text{seg}} = 400$, 9 system qubits) on a CPU statevector simulator. All segment construction and metrics use the optimised **SimpleHamiltonianFast** and Numba **fast_segment_metrics** code paths. The full multiplicity landscape ($\theta_d \times \text{noise} \times T$ up to 1000, quantum for $T \leq 200$) is being produced on HTCondor and is not yet included here.

4 Results

4.1 Hamiltonian structure and eigenspectrum

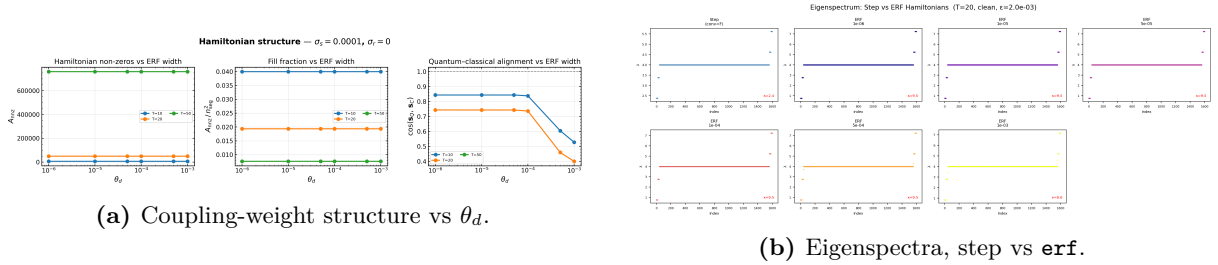


Figure 1: Structure of the compatibility matrix. The step kernel produces binary couplings; the **erf** kernel spreads weight smoothly across all mid-hit pairs, yielding a denser matrix and a compressed eigenspectrum.

The condition number $\kappa = \lambda_{\text{max}}/\lambda_{\text{min}}$ (Table 1) is $\kappa_{\text{step}} = 2.36$ and $\kappa_{\text{erf}} = 9.47$ for $\theta_d \leq 5 \times 10^{-4}$, falling to 8.55 at $\theta_d = 10^{-3}$ as the widest kernel weakens the strong true-true couplings and compresses the spectrum. Crucially κ is *unchanged* by the 1% hit drop (Fig. 2): the spectral shape is a property of θ_d and the geometry, not of the noise realisation.

Table 1: Matrix condition number κ at $T = 20$ ($N_{\text{seg}} = 1600$). Identical for the clean and 1%-drop events.

	step	$\theta_d=10^{-6}$	10^{-5}	5×10^{-5}	10^{-4}	5×10^{-4}	10^{-3}
κ	2.36	9.47	9.47	9.47	9.47	9.47	8.55

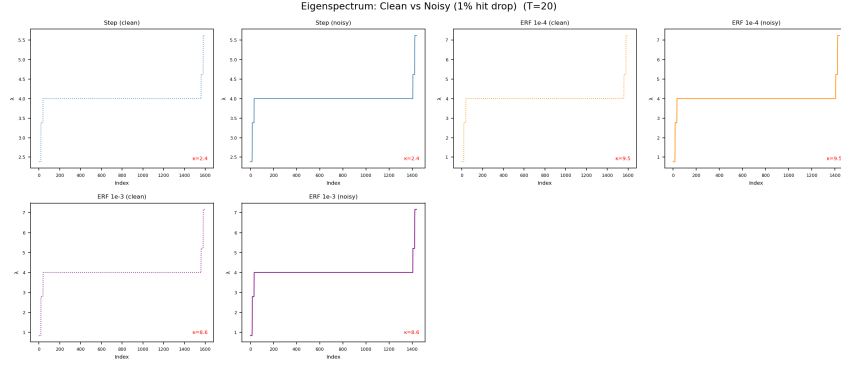
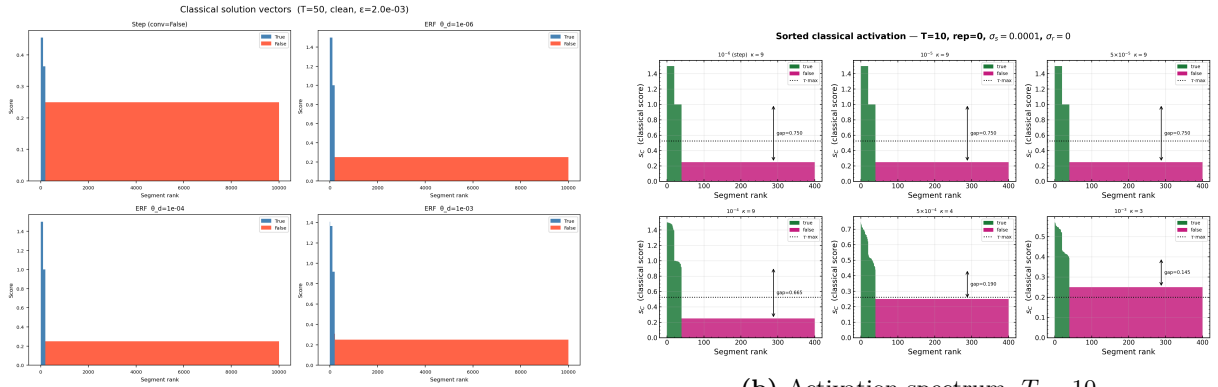


Figure 2: Eigenspectrum change under 1% hit drop. Both kernels are spectrally noise-invariant; κ is identical clean vs noisy for every θ_d .

4.2 Classical solution vectors



(a) Sorted classical activations.

(b) Activation spectrum, $T = 10$.

Figure 3: Classical solution vectors. Step-function true segments sit on the 0.375 plateau with false segments at the 0.25 attractor; the **erf** kernel pushes true segments higher (towards 1.0–1.5) but also lifts a few borderline-false pairs towards 0.5.

Under the step kernel the true activations occupy a narrow 0.36–0.45 band and the false activations the 0.25–0.33 band, perfectly separated by the 0.35 threshold. The **erf** kernel widens the true band upward (stronger coupling) but, because it gives partial weight to triplets just outside ε , a small number of false segments are lifted toward 0.5 — the origin of the **erf** false-rate penalty quantified below.

4.3 Segment metrics: clean and noisy

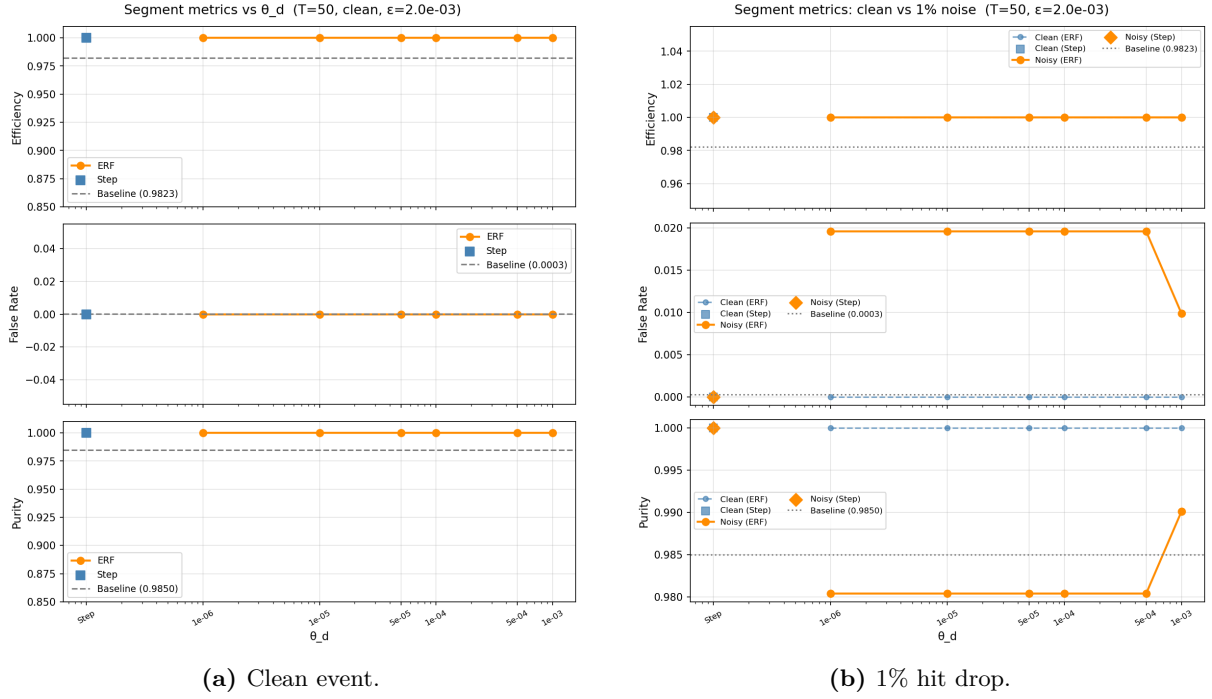


Figure 4: Segment efficiency, false rate and purity vs θ_d at $T = 50$, at the absolute threshold $x > 0.35$. Step shown as the leftmost anchor.

On the clean event every kernel achieves efficiency = 1.000, false rate = 0.000 and purity = 1.000 (Table 2). Under 1% hit drop the step kernel remains perfect, whereas the `erf` kernels acquire a small false rate of 0.0196 ($\theta_d \leq 5 \times 10^{-4}$) or 0.0099 ($\theta_d = 10^{-3}$), i.e. purity 0.980–0.990. In other words, in the low-noise regime the simpler step function is at least as good and carries no false-acceptance cost.

Table 2: Segment metrics at $T = 50$ (threshold $x > 0.35$).

Kernel	Clean			1% hit drop		
	eff.	false	purity	eff.	false	purity
step	1.000	0.000	1.000	1.000	0.000	1.000
<code>erf</code> $\theta_d \leq 5 \times 10^{-4}$	1.000	0.000	1.000	1.000	0.0196	0.980
<code>erf</code> $\theta_d = 10^{-3}$	1.000	0.000	1.000	1.000	0.0099	0.990

4.4 Resolution scan: where erf wins

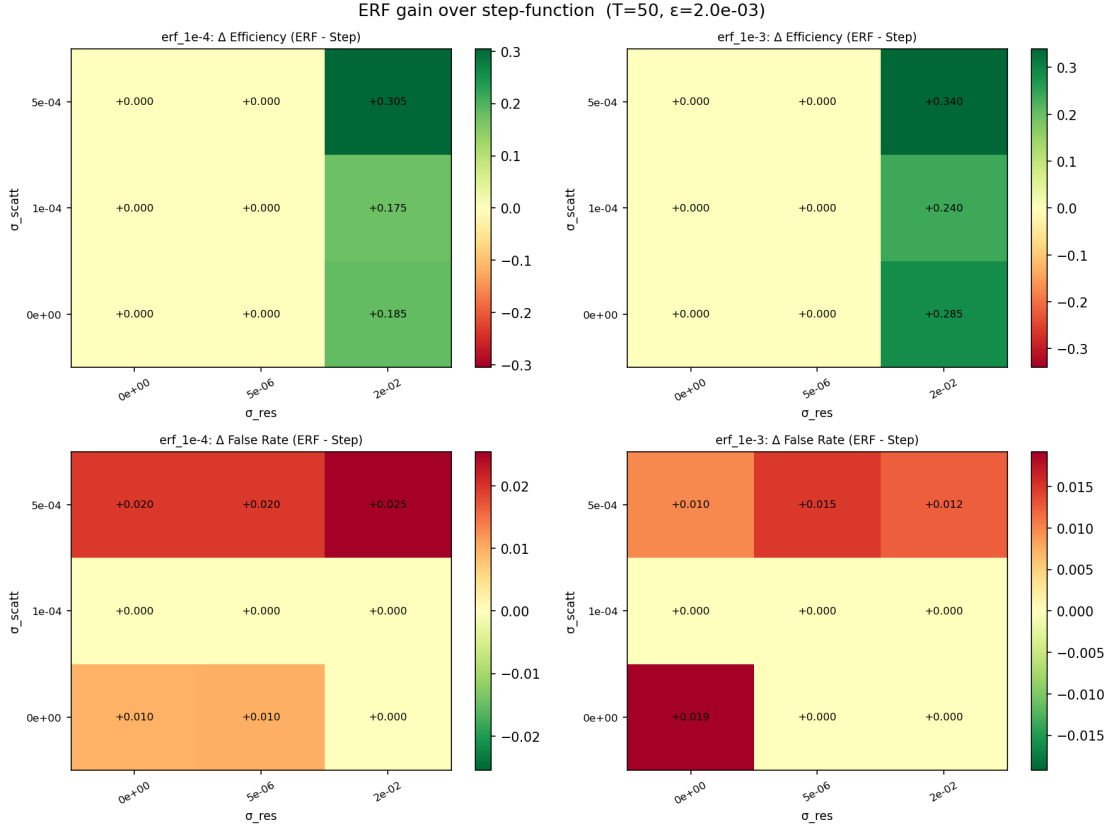


Figure 5: Segment efficiency vs noise condition for step, $\text{erf } \theta_d=10^{-4}$ and $\text{erf } \theta_d=10^{-3}$, over the $(\sigma_{\text{scatt}}, \sigma_{\text{res}})$ grid at $T = 50$. The **erf** kernel recovers a large fraction of efficiency at high measurement resolution where the step kernel collapses.

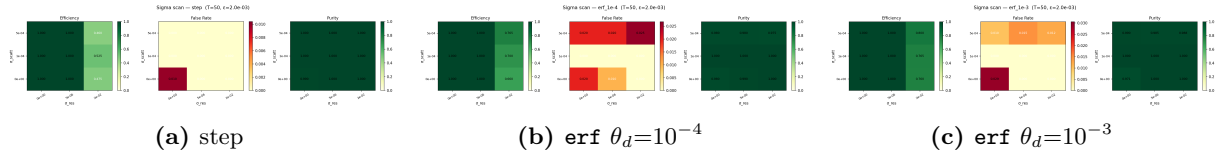


Figure 6: Efficiency heatmaps over $(\sigma_{\text{scatt}}, \sigma_{\text{res}})$. The bottom row ($\sigma_{\text{res}} = 0.02$) is where the kernels diverge.

The scan (Table 3) isolates the regime where the smooth kernel matters. At low resolution ($\sigma_{\text{res}} \leq 5 \times 10^{-6}$ mm) all kernels give efficiency = 1.000 regardless of σ_{scatt} . At $\sigma_{\text{res}} = 0.02$ mm the step efficiency collapses to 0.46–0.53, because the hard cut rejects true triplets whose kink angle has been smeared past ϵ ; the **erf** kernel, which still assigns them partial weight, recovers efficiency to 0.66–0.80, with the widest kernel ($\theta_d = 10^{-3}$) best at 0.76–0.80. The recovery is +0.27 to +0.34 in absolute efficiency, at the cost of a small false rate (0.012–0.026).

Table 3: Segment efficiency at $T = 50$ across the noise scan (threshold $x > 0.35$). Only the $\sigma_{\text{res}} = 0.02$ rows distinguish the kernels.

σ_{scatt}	σ_{res} (mm)	step	$\text{erf } \theta_d=10^{-4}$	$\text{erf } \theta_d=10^{-3}$
0	0.02	0.475	0.660	0.760
10^{-4}	0.02	0.525	0.700	0.765
5×10^{-4}	0.02	0.460	0.765	0.800
$\sigma_{\text{res}} \leq 5 \times 10^{-6}$ (any σ_{scatt})		1.000	1.000	1.000

4.5 Quantum solution comparison

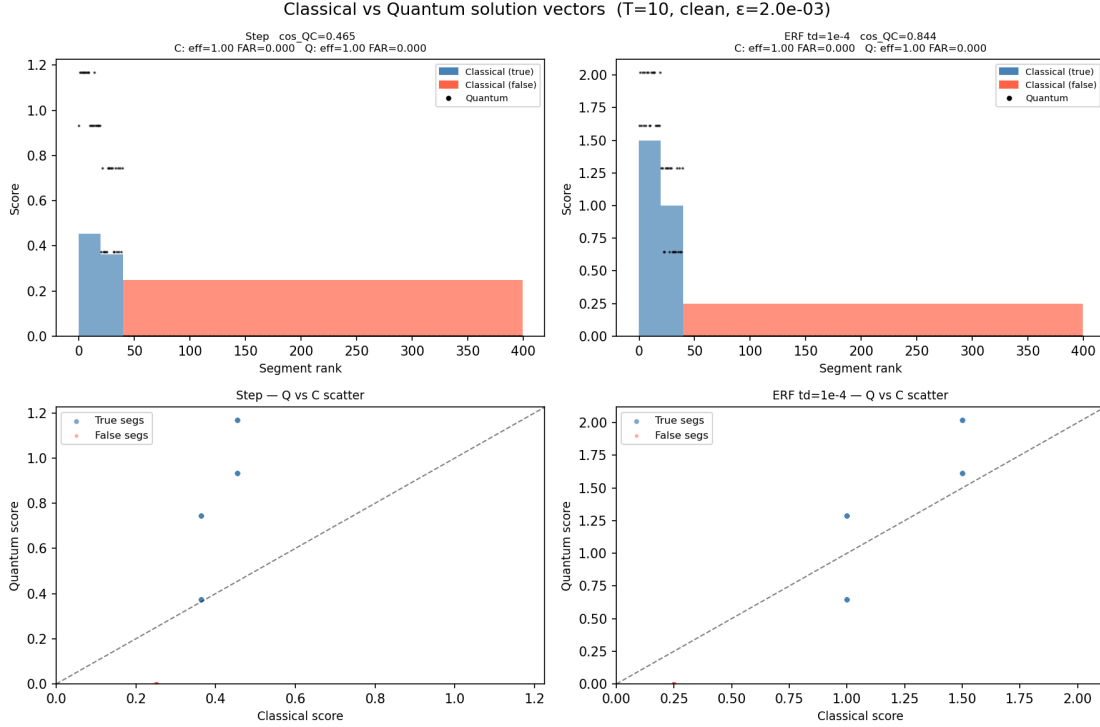


Figure 7: Classical vs 1BQF quantum solution vectors at $T = 10$ ($N_{\text{seg}} = 400$, 9 system qubits), for the step and **erf** $\theta_d = 10^{-4}$ Hamiltonians. Left: sorted classical activations with the quantum solution overlaid; right: quantum-vs-classical scatter.

Both kernels classify every segment correctly at the absolute threshold (efficiency = 1.000, false rate = 0.000 for classical and quantum alike at $T = 10$). The *fidelity* of the quantum solution to the classical one, however, differs sharply: the cosine similarity is $\cos \theta_{QC} = 0.465$ for the step kernel but 0.844 for **erf** $\theta_d = 10^{-4}$, with identical ancilla success probability $P_{\text{anc}} = 0.0224$. The smoother **erf** spectrum aligns the HHL-prepared state with the classical solution far better, even though its condition number is larger ($\kappa_{\text{erf}} = 9.5$ vs $\kappa_{\text{step}} = 2.4$). This is the strongest standalone argument for the **erf** kernel.

5 Discussion

When does the **erf kernel help?** The smooth kernel is not a uniform improvement. In low-to-moderate noise the step function is already perfect at the segment level and the **erf** kernel only adds a small false-acceptance penalty by weighting borderline-false pairs. The **erf** kernel pays off in exactly one classical regime: large measurement resolution ($\sigma_{\text{res}} \sim 0.02$ mm), where its tolerance of smeared true triplets recovers +0.15 to +0.34 absolute efficiency that the hard cut discards. The widest kernel tested, $\theta_d = 10^{-3}$, is consistently best for this classical-efficiency recovery.

The condition-number / fidelity puzzle. Naively, a smaller condition number should make the quantum (HHL) solution more faithful. The data show the opposite: the step matrix is better conditioned ($\kappa = 2.4$) yet has the *worse* quantum fidelity ($\cos \theta_{QC} = 0.47$), while the **erf** matrix ($\kappa = 9.5$) reaches $\cos \theta_{QC} = 0.84$. The resolution is that the 1-bit HHL fidelity is governed by how well the eigenvalue *distribution* matches the single phase bit, not by the extremal ratio κ : the **erf** kernel’s smoother, more clustered spectrum is better matched by the 1-bit filter. The

quantum benefit of **erf** is therefore genuine and independent of the classical-efficiency story.

Trade-off summary. The **erf** kernel costs a $74\times$ denser matrix and a small ($\leq 3\%$) false-rate penalty in noisy conditions, and buys (i) classical robustness to resolution smearing and (ii) roughly doubled quantum fidelity. Whether the density cost matters at scale, and whether the fidelity gain persists to higher multiplicities, is the subject of the in-progress HTCondor sweep.

6 Conclusions

1. **The step function is the right default at low noise.** With the absolute threshold $x > 0.35$ it is perfect (efficiency 1.000, false rate 0.000); the **erf** kernel matches it but adds a $\leq 2\%$ false rate under hit drop.
2. **erf wins under heavy resolution smearing.** At $\sigma_{\text{res}} = 0.02$ mm it recovers efficiency from ~ 0.5 (step) to $0.76\text{--}0.80$ ($\theta_d = 10^{-3}$), a $+0.27$ to $+0.34$ absolute gain at a small false-rate cost.
3. **erf roughly doubles quantum fidelity.** $\cos\theta_{QC}$ rises from 0.47 (step) to 0.84 (**erf**, $\theta_d = 10^{-4}$) at $T = 10$, despite a larger κ ; the 1-bit HHL filter is matched to the spectral shape, not to κ .
4. **The spectrum is noise-invariant.** κ depends only on θ_d (2.4 step, $8.6\text{--}9.5$ **erf**) and is identical clean vs 1% -drop.
5. **Recommended operating points.** Use the step kernel by default; switch to **erf** $\theta_d = 10^{-3}$ when measurement resolution is large, or **erf** $\theta_d = 10^{-4}$ when running the quantum pipeline.
6. **Follow-up (in progress).** The HTCondor sweep ($\theta_d \times (\sigma_{\text{scatt}}, \sigma_{\text{res}}) \times T$, 882 jobs, quantum for $T \leq 200$) will extend these single-event findings to the full multiplicity landscape and supply the Youden-J / EER threshold analysis.

References

- [1] X. Chiotopoulos, D. Nicotra, G. Scriven, K. Driessens, M. Merk, J. Schütz, J. de Vries, M. H. M. Winands, *TrackHHL: The 1-Bit Quantum Filter for particle trajectory reconstruction*, arXiv:2601.07766 [quant-ph] (2026).
- [2] A. W. Harrow, A. Hassidim, S. Lloyd, *Quantum Algorithm for Linear Systems of Equations*, Phys. Rev. Lett. **103**, 150502 (2009).